

Computer Workshop 6: Diagnosing linear models with a categorical predictor

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Workshop Structure

As always, we are using the workflow below to structure the workshops:

1. Formulate a research question.
2. Perform exploratory data analysis.
3. Identify any hidden assumptions.
4. Fit an appropriate model.
5. Diagnose the model and check assumptions [Focus of today].
6. Summarise the results.
7. Interpret and provide inferences.

For this workshop, we'll continue with the models you ran on Tuesday, going back to step 5 to complete our workflow.

If you didn't manage to finish Workshop 5, please do so now before moving on to diagnosing your model.

A Reminder of the Model

On Tuesday you ran a model attempting to understand if Java mongoose may be decreasing the average weight of Amami rabbits on two islands, south of mainland Japan. Our model took the form:

$$weight_i \sim Normal(\mu_i, \sigma)$$

$$\mu_i = \beta_0 + \beta_1 \times Present_i$$

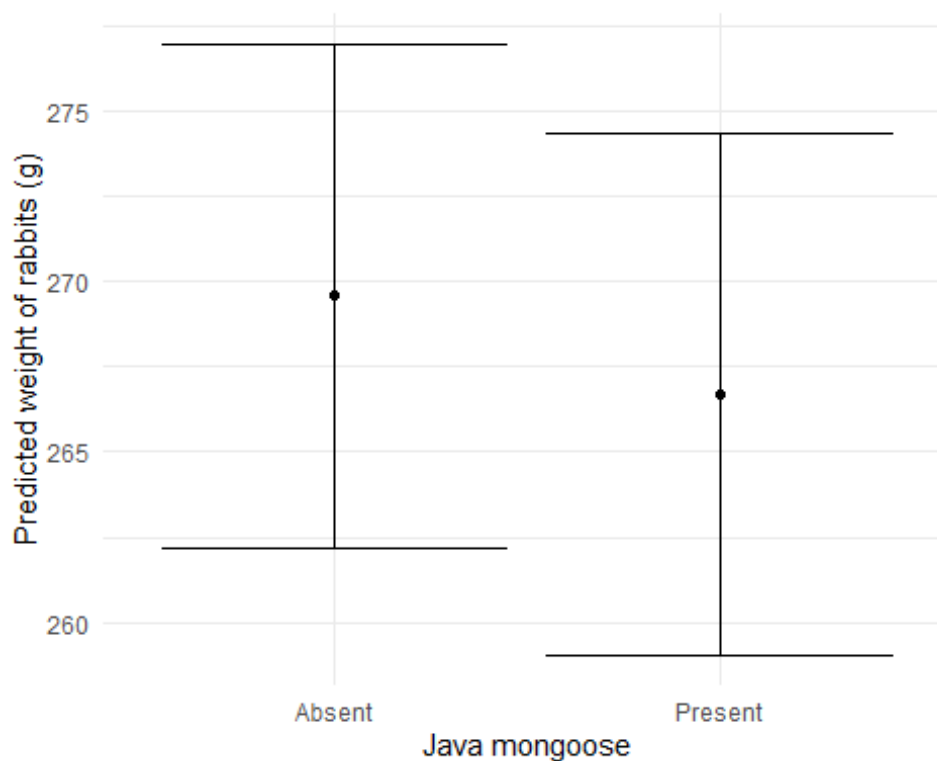
where β_0 was the average weight of rabbits when mongoose were absent (i.e. when $Present = 0$), and β_1 was the difference from that average weight when mongoose were present (i.e. when $Present = 1$).

Having fit the model, we can replace the two, previously unknown, parameters β_0 and β_1 with what we now estimate these values to be:

$$\mu_i = 269.6 + -2.9 \times Present_i$$

If you get different parameter estimates, you've likely missed one or both data entry mistakes present in the data. Carry out an EDA to identify and resolve these before running the model.

We then used that model to create predictions and plotted these:

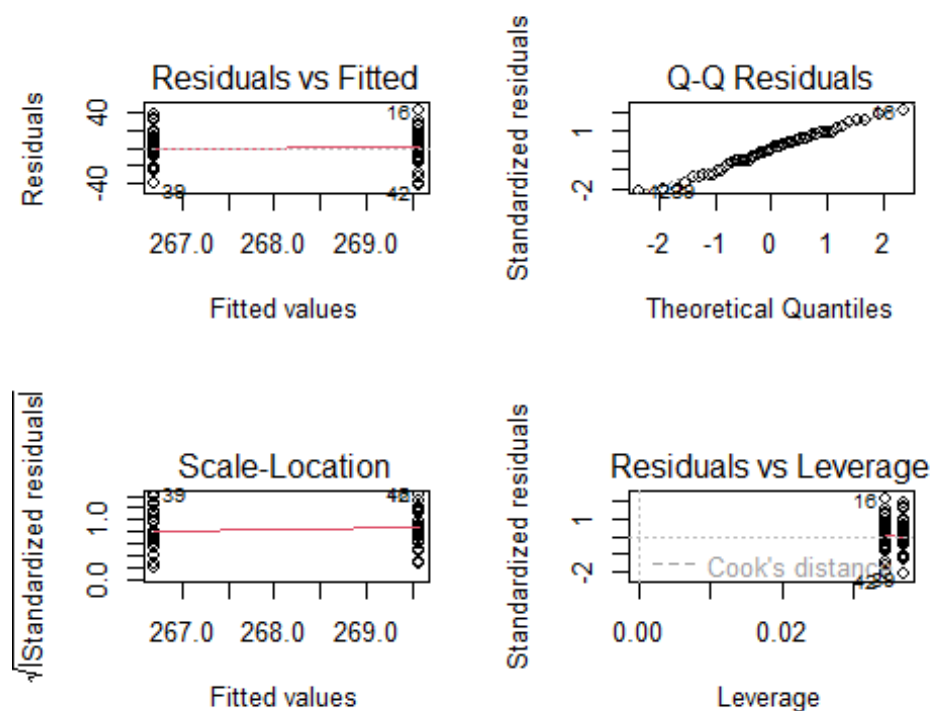


But, as ever, can we convince ourselves that this model is worthy of our trust before we present it to others?

Diagnosing

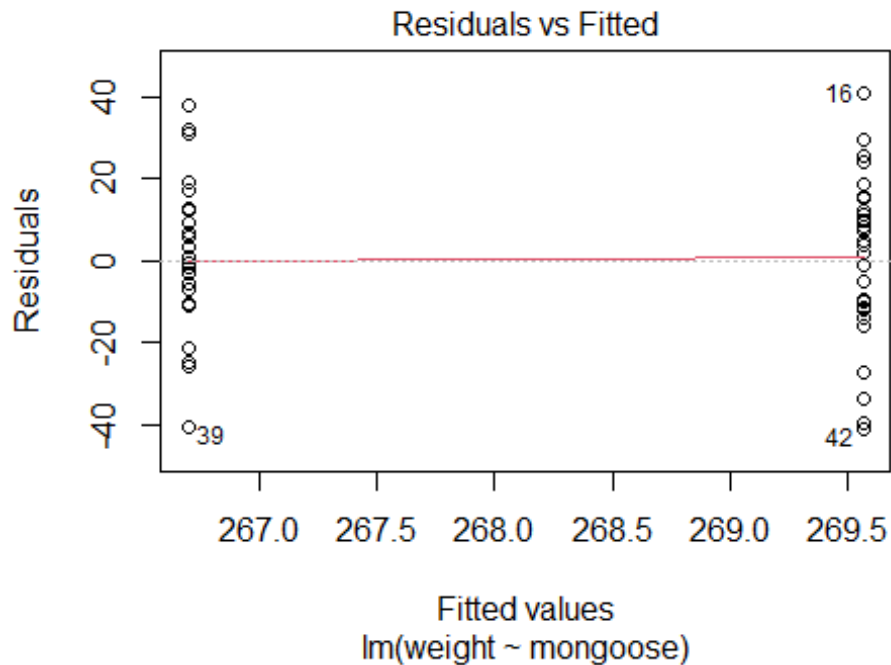
The process of diagnosing is exactly the same as with linear models with a continuous predictor, the only difference coming in one plot. Let's go through these plots to remind ourselves what we're looking for, how important it is to meet a respective assumption, and whether that should influence how we see the model.

```
par(mfrow = c(2,2)) # split figure into a 2 x 2
plot(rabbit_mod)
```



```
par(mfrow = c(1,1)) # reset to 1 x 1
```

Residuals vs Fitted



As a reminder, the y-axis is residual errors (the difference between an actual measurement and the prediction for that same measurement).

The x-axis represents the **Fitted values**, which is just a different way of saying “Predicted values”. Here, our fitted values are the predicted weight of rabbits on either island with or without mongoose.

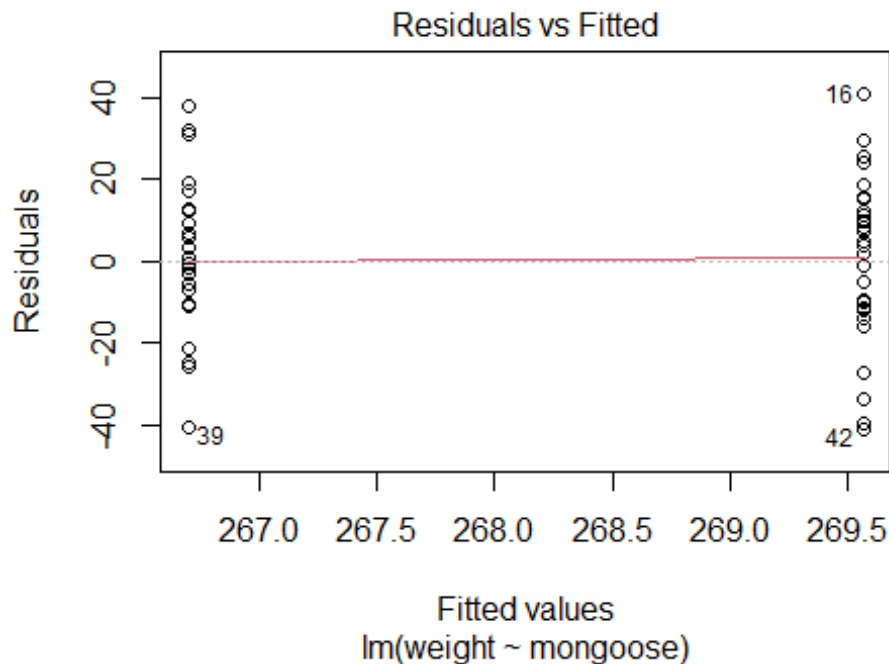
Linearity

One of the main differences from last week is that our model doesn't have a linear trend - there's no line that we can draw between mongoose being present and mongoose being absent. So we're actually not assuming linearity here. One less thing to check for :).

Equal Variance of Errors

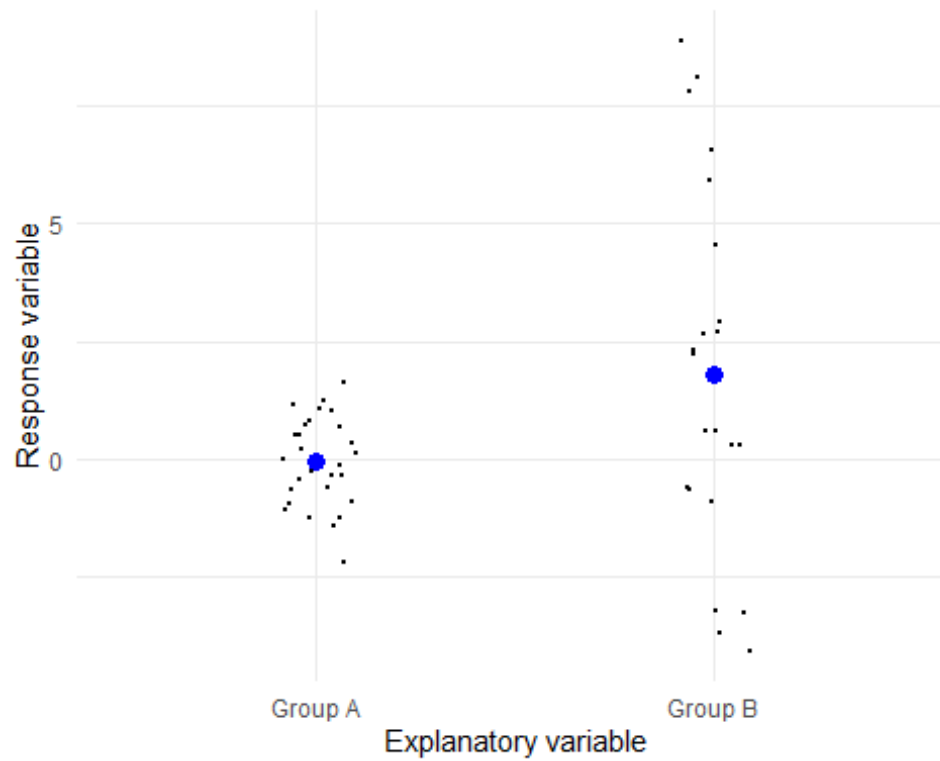
We can, however, check for equal variance of errors.

```
plot(rabbit_mod, which = 1)
```

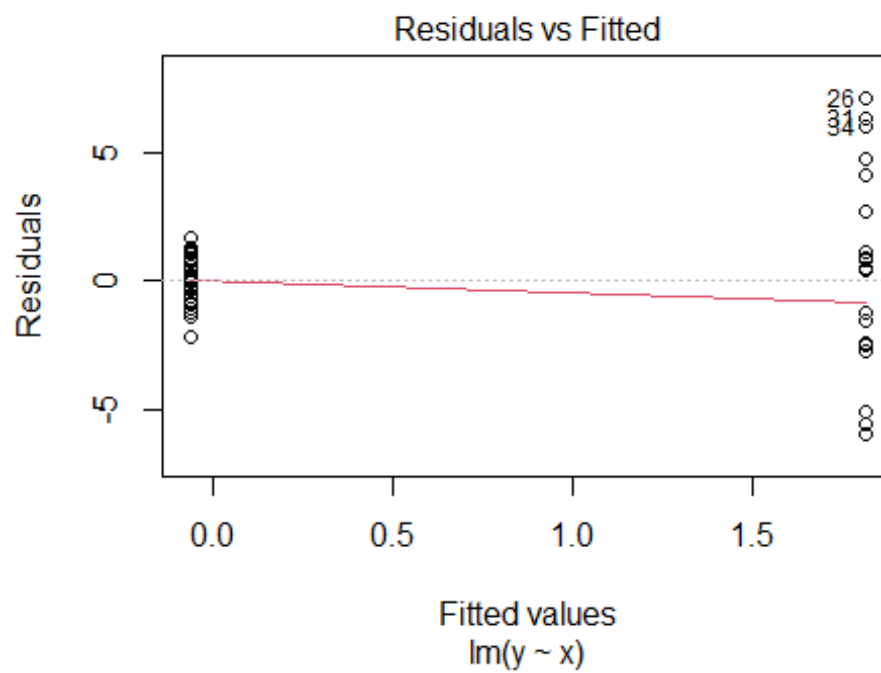


What we're looking for here is whether each category (here where mongoose are Present or Absent) has broadly equal spread above and below a residual error of zero. The two lines you see here are the two groups. Remember that there was about a five gram difference between the island with mongoose and the one without, and we can see that here - when mongoose are absent the model predicts they should weigh about 269 g and when there are mongoose the model predicts rabbits weigh 266 g. In both groups, there are rabbits that weigh about 40 g more or less than the model expects, so it looks pretty good for meeting our assumption of equal variance of errors.

What would a "bad" example look like? Well, consider the following example using a simulated dataset where the blue dots show the predicted response variable for groups A and B:

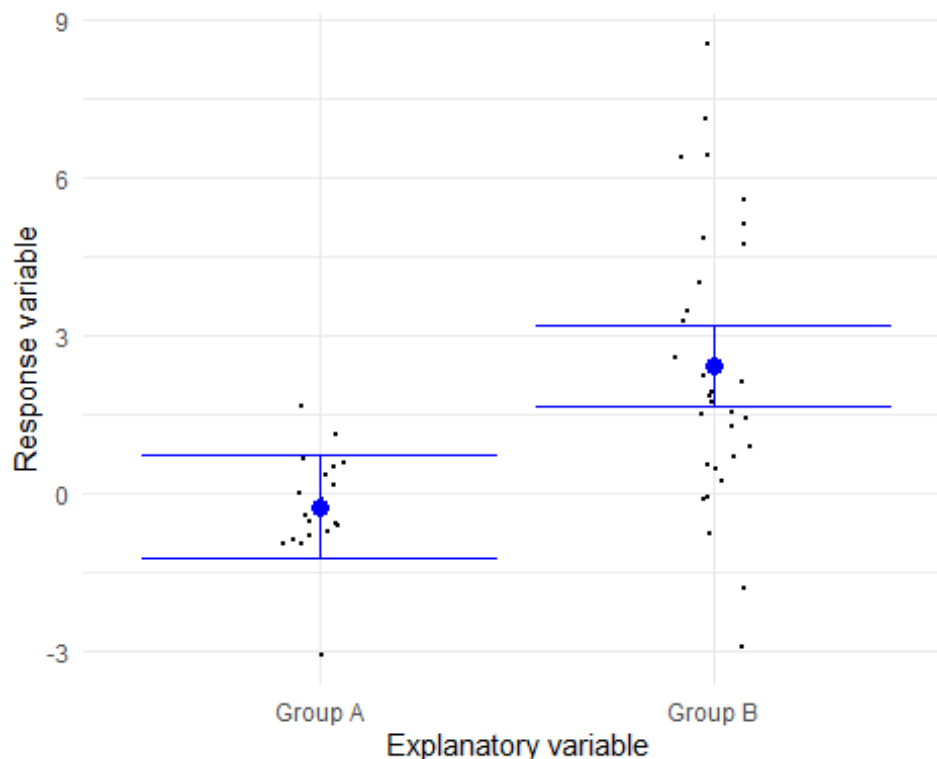


When we look at the Residuals vs Fitted plot, we can see:



The amount of variation in the error (or residuals) is not equal here. When the predicted values (x-axis) are < 0 , we see very little variation around those predictions (residuals within ca. 2 units from the observed value), while at the other end of the figure (corresponding to Group B), we can see a lot more variation in errors, up to 8 units from the true value.

When we add in the 95% confidence intervals to the plot:

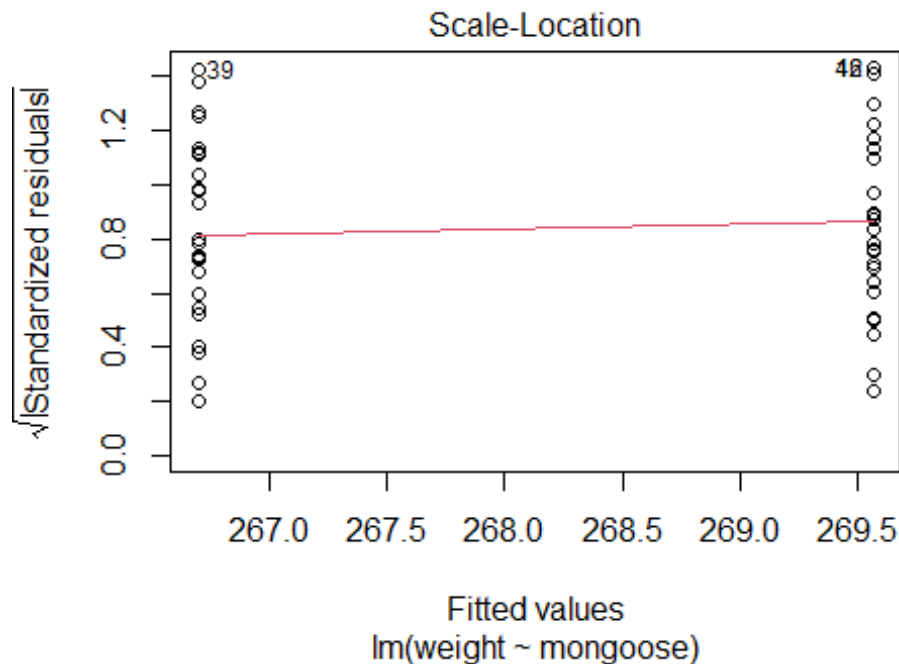


We can see that the amount of uncertainty (the 95% CI) is equal in both groups, but in reality, the confidence intervals for Group A should probably be much smaller. Keep in mind that inflated standard errors (which we use to calculate 95% CI) are a consequence of violating the assumption of equal variance of errors. However, the point estimate (the mean of groups A and B) are estimated accurately (which I only know for sure in this case because I simulated the data). So if we only cared about estimating the means of the groups accurately (and are not too concerned about uncertainty), then we wouldn't be too worried about having unequal variance of errors.

It's worth saying that we generally do care about estimating uncertainty accurately. Keep in mind, we never really know if the means are "True" (as in they accurately reflect reality), so having some measure of uncertainty is generally the responsible approach to science. If you need a reminder of why it's important, have a look at the lecture where we spoke about being given the option of two drugs with different probabilities of unintended mortality, and how our decision would change depending on if we showed uncertainty or not.

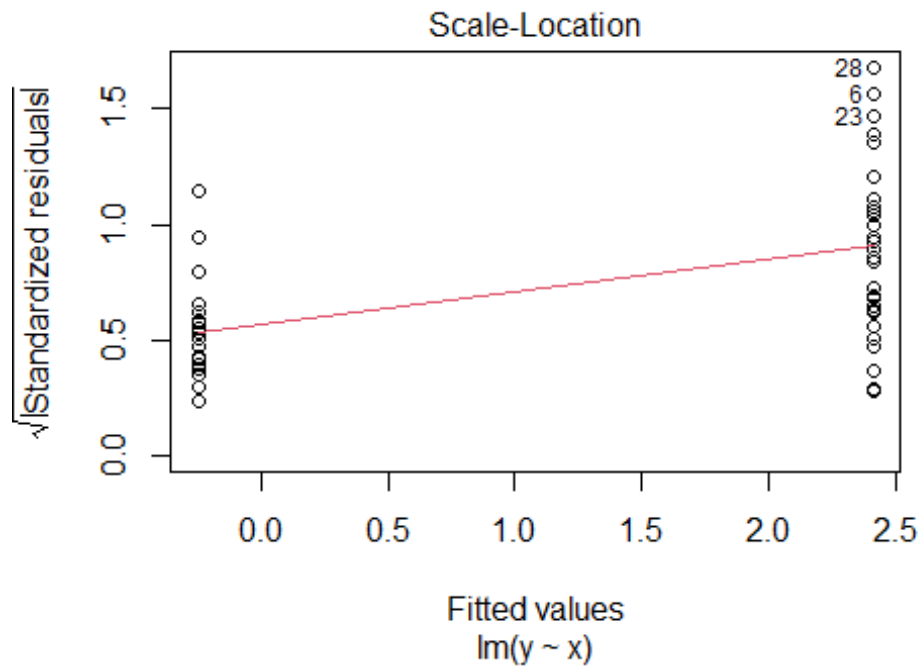
Scale-Location

Remember that the Scale-Location plot offers us another chance to evaluate the equal variance assumption.



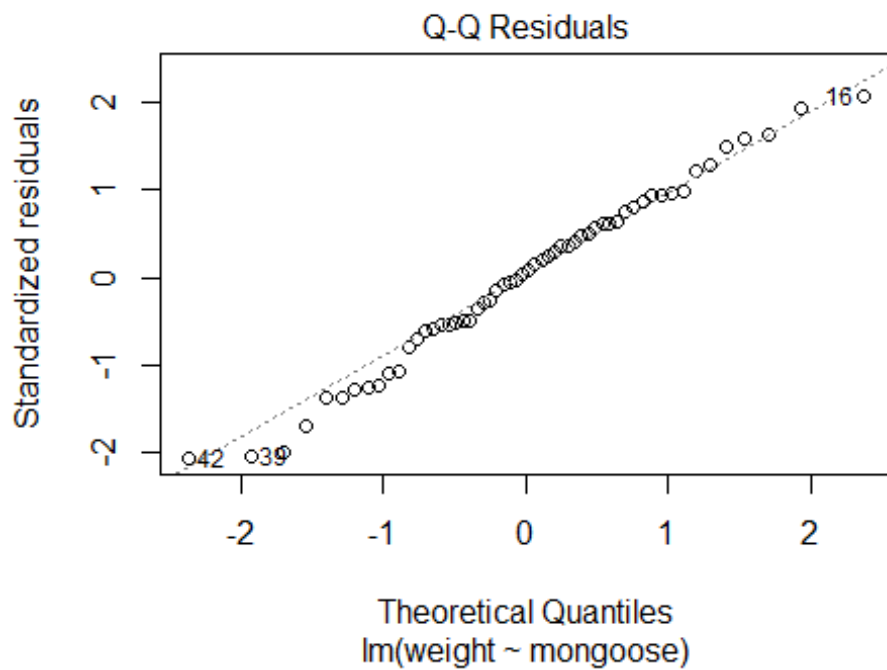
From the plot above, it seems like the assumption is very slightly less well met than the Residuals vs Fitted plot would suggest, simply because predictions at about 270 g seem to have residuals that are the tiniest bit larger than when the rabbits are predicted to weigh roughly 265 g (representing the mean predicted weights for our two mongoose categories, Present and Absent respectively). Importantly though, this is no where near “substantial” enough for me to be worried about.

For example, compare it to the data I simulated above that includes unequal variance of errors by design. Here’s what the Scale-Location plot looks like in that case:



Q-Q Residuals

Next to the Q-Q plot for the rabbit model.

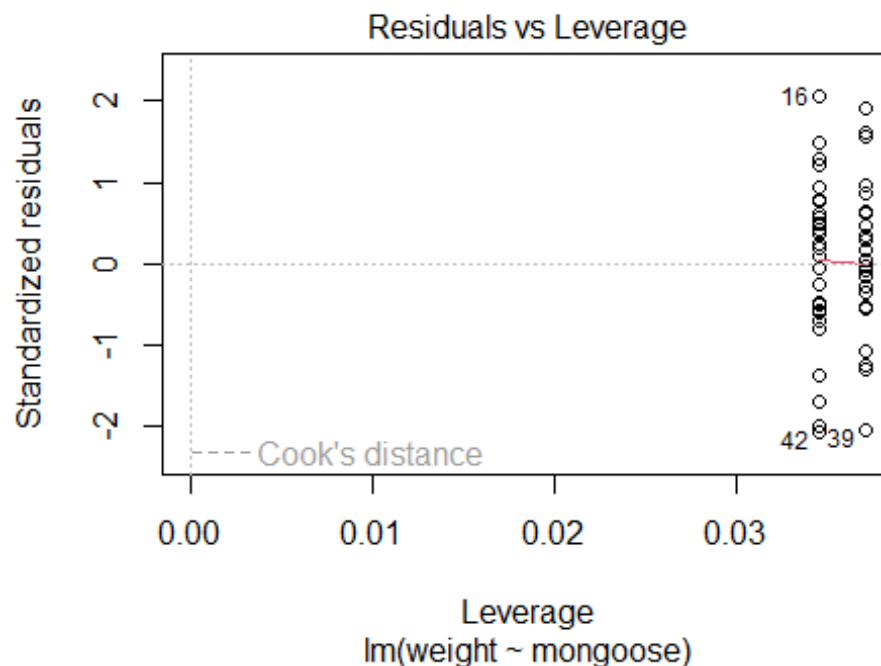


Keep in mind that we're assuming that the residuals of our model take on a *Normal* distribution, so that's what this plot is looking for. What having *Normally* distributed residuals actually means, though, is whether or not our model is predicting the individual rabbits of our dataset accurately, which 99% of the time, we're not too bothered about.

Never-the-less, we still want the points to broadly lie on the dashed line, but we don't stress out too much if it's not exactly perfect. If we ever did see utterly bizarre patterns in the points, that implies one of two things. Either you need to choose a different distribution for your data generating process and fit a "Generalised Linear Model" (GLM), or your model is missing some important predictor variables (more on both of these options later in the course).

Residuals vs Leverage

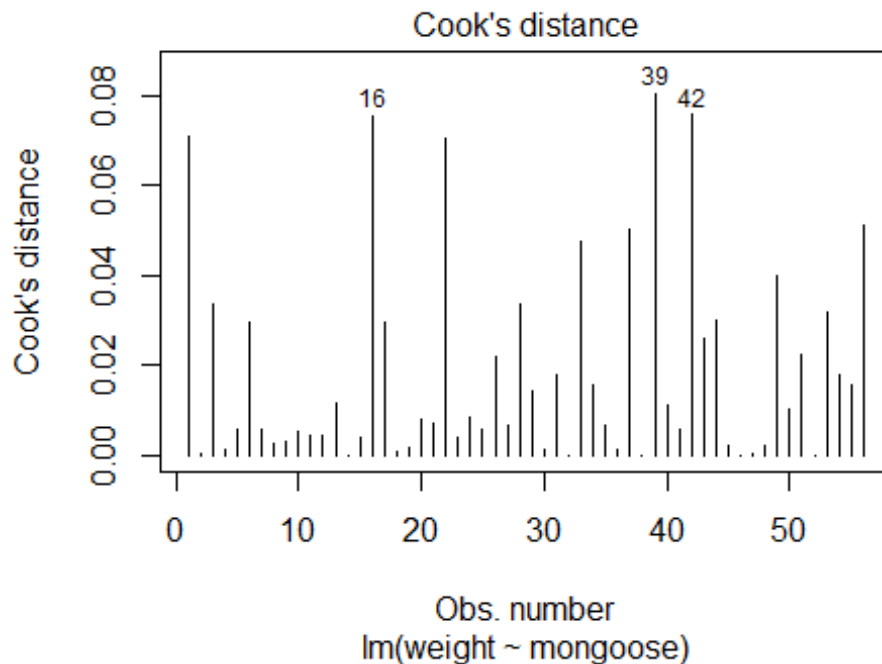
The last plot doesn't assess the assumptions we've discussed but provides a different kind of insight. It helps identify disproportionately influential data points.



Just like last week, we're looking to see if we can spot the contour lines indicating a Cook's distance of 1. In the plot above, these contour lines are no-where in sight, so we don't have any issues here either.

If you want an alternative way to check Cook's distance, without showing Leverage, then you can do:

```
plot(rabbit_mod, which = 4)
```



None of our points are close to 1, so we don't have any observations that are pulling our line one way or another.

Question Set One

1. Which assumption(s) can be checked for using the Residuals vs Fitted plot when fitting a linear model with only a categorical predictor variable?

Equal Variance of Errors. We have no linear trend, so there's no assumption of linearity.

2. Which assumption(s) can be check for using the Scale-Location plot when fitting a linear model with only a categorical predictor variable?

Equal Variance of Errors

3. Which assumption(s) can be check for using the Q-Q Residuals plot?

Normality of Errors

4. Is meeting the assumption of Equal Variance of Errors important?

Having unequal variance errors (or residuals) can inflate the standard error. This means your 95% confidence intervals will be inflated which can make drawing inference a bit more challenging. It's a bigger issue when using P-Values and Null Hypothesis Significance Testing, where it leads to inflated type 1 and type 2 errors (thinking something is significant when it's not, or vice versa)

5. Using the diagnostics from the Amami rabbit model, are there any diagnostic plots which indicate a problem in the model?

```
# No, they seem pretty well met across the board.
```

6. Does meeting the assumptions of these plots mean the model is trustworthy?

```
# No, it just means we have not found any issues related to a subset of our assumptions. Keep in mind that on Tuesday we were concerned about the validity of mongoose being absent on one of the islands. That's the biggest hurdle for this model, and these diagnostic plots don't help us at all with that.
```

A More Visually Appealing Alternative

If you preferred the plots made by performance using `check_model()` you can use them for the rabbit model, but note that it doesn't work particularly well for our simple model. For instance, it tries to draw a green line between our two groups, which looks pretty wiggly, but that doesn't actually mean anything here.

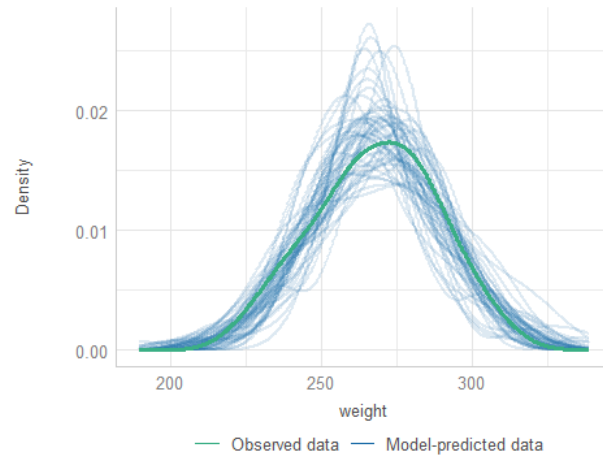
```
library(performance)
```

```
## Warning: package 'performance' was built under R version 4.3.3
```

```
check_model(rabbit_mod)
```

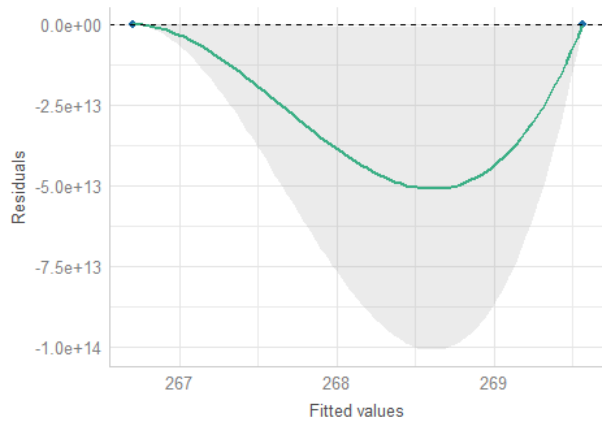
Posterior Predictive Check

Model-predicted lines should resemble observed data line



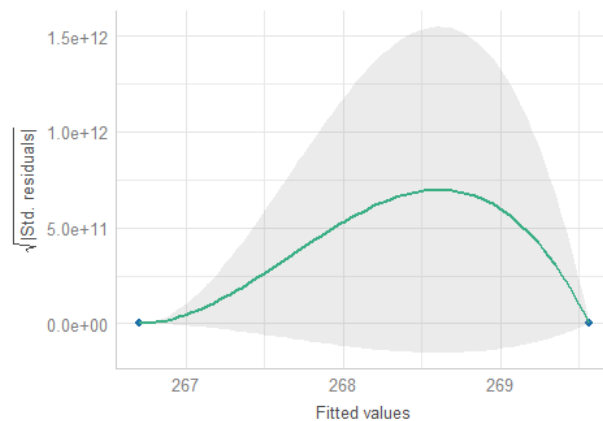
Linearity

Reference line should be flat and horizontal



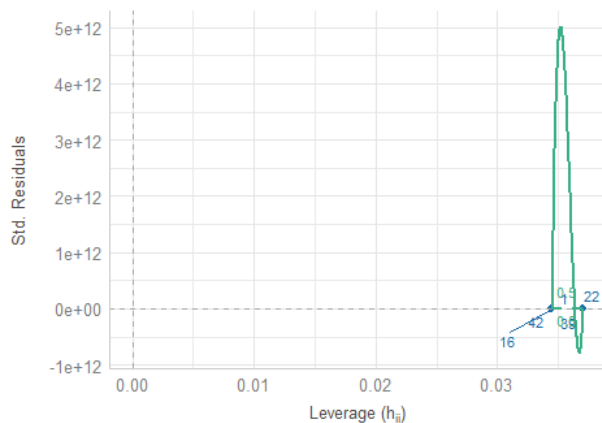
Homogeneity of Variance

Reference line should be flat and horizontal



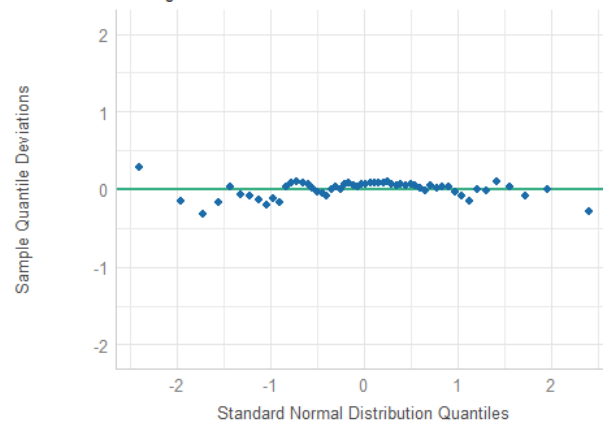
Influential Observations

Points should be inside the contour lines



Normality of Residuals

Dots should fall along the line



The Lemurs

Having diagnosed the Amami rabbit model, let's move on to the lemur model from the last workshop. If you haven't run that model yet, do so now. Once you have, go through the process

of checking the model diagnostics to determine if you're willing to place your trust in the model.

```
# Load data
lemur <- read.table("H:/BI3010/Workshop3/lemur.txt", header = TRUE)

# This is how I would analyse the Lemur data

# Deon Roos
# Analysis of lemur activity by habitat type
# In which habitat are lemurs most active

# Packages -----
library(ggplot2)    # For plotting
library(ggeffects)  # For quick plotting of model fit

# Data -----
# (Not running these here because of how this document is created)
# setwd(H:/BI3010/)
# lemurs <- read.table("lemurs.txt", header = TRUE)

# EDA -----

str(lemur)
# n = 120
summary(lemur)
# No apparent issues

ggplot(lemur) +
  geom_histogram(aes(x = activity), colour = "white", binwidth = 0.5) +
  theme_minimal()

ggplot(lemur) +
  geom_histogram(aes(x = activity, fill = habitat),
    colour = "white", binwidth = 0.5) +
  theme_minimal()
# Data entry mistake spotted
# One habitat is recorded as "Rain forest" and not "Rainforest"
# Obviously a mistake, so happy to correct without confirmation

# Correcting data entry mistake -----
lemur$habitat[lemur$habitat == "Rain forest"] <- "Rainforest"

# Check if correction worked
ggplot(lemur) +
  geom_histogram(aes(x = activity, fill = habitat),
    colour = "white", binwidth = 0.5) +
  theme_minimal()

# Hidden assumption -----
```

```

# - Validity - Unsure how activity is measured but must be hard. Seems possible that accuracy changes in different
# habitats? E.g. easier to keep track of lemurs in dry forest than rain forest?
# - Equal variance? - Seems likely that some amount of variation may differ between habitats? Something to check.

# Model fit -----
lem_mod <- lm(activity ~ habitat, data = lemur)

# Diagnose -----
par(mfrow = c(2,2))
plot(lem_mod)
par(mfrow = c(1,1))
# No blatant issues from diagnostics. Errors seem fairly equal.

# Summarise -----
summary(lem_mod)
# Model predicts 7.9 hours of activity in Dry forests
# 1.05 hours *less* activity in Mangroves (so 6.85 hours total)
# 1.8 hours *more* activity in Rainforests (so 9.7 hours total)
# 1.8 hours *less* activity in Spiny Forests (so 6.1 hours total)

preds <- ggpredict(lem_mod)
plot(preds)

# Interpret -----
# No issues identified in EDA. Lemurs most active in rain forests and least in spiny forests. Could be interpreted as meaning rainforests are the preferred habitat, but could also imply this is just where they are most active (e.g. spend more time foraging for equivalent amount of food). Potential that this may be biased by how easy it is to record activity in different habitats but cannot confirm with data as is.

```

Assumption table

Assumption	Description	Example of Violation
1. Validity	The data is able to answer the research question effectively.	Using IQ to measure overall intelligence (what does IQ even measure?).
2. Representativeness	The data reflects the population or group being studied.	Using survey data from a rural area to draw conclusions about an urban population.
3. Additivity	Effects of predictors are additive and there are no interactions between them.	Assuming that study hours and sleep hours independently affect exam scores without considering their combined effect (imagine a

Assumption	Description	Example of Violation
		Venn diagram with study and sleep overlapping).
4. Linearity	The relationship between predictors and the outcome is linear.	Assuming plant growth will continue to increase indefinitely without ever levelling out.
5. Independence of Errors	Observations are independent of each other. (Related to residual errors.)	If modelling height, the data come from the same group of children over time.
6. Equal Variance of Errors	Residuals have constant variance across all levels of predictors ("homoscedasticity"). (Related to residual errors.)	If modelling income, variability in salaries is not the same across all levels of a company's hierarchy.
7. Normality of Errors	Residuals should be normally distributed. (Related to residual errors.)	If modelling income, the presence of any extremely wealthy individuals will be hard for the model to understand and explain.